

Notes: Nickell (Econometrica, 1981)

Biases in Dynamic Models with Fixed Effects

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1 Big picture

- **Question.** What happens when we estimate a dynamic panel model with individual fixed effects by the standard within-group estimator when the number of individuals N is large but the number of time periods T is fixed?
- **Core answer.** The within estimator is inconsistent in dynamic fixed-effects panels with fixed T . Even as $N \rightarrow \infty$, the demeaned lagged dependent variable remains correlated with the demeaned error term.
- **Name of the problem.** This is the classic **Nickell bias**. The paper derives analytical expressions for the asymptotic bias in first-order autoregressive panel models with fixed effects.
- **Why fixed effects matter.** Fixed effects are attractive when unobserved individual heterogeneity is correlated with regressors. But in a dynamic panel, the same demeaning operation that removes the fixed effect mechanically creates endogeneity between the lagged dependent variable and the transformed error.
- **Main result.** In the simple AR(1) fixed-effects model, the bias in the coefficient on $y_{i,t-1}$ is negative for positive persistence ρ . For moderate T , it is large; for large T , it is approximately

$$\text{plim}_{N \rightarrow \infty}(\hat{\rho} - \rho) \approx -\frac{1 + \rho}{T - 1}.$$

- **Example.** If $T = 10$ and $\rho = 0.5$, the asymptotic bias is about -0.162 , so the fixed-effects estimate converges to roughly 0.338 rather than 0.5 .
- **Important detail.** The bias does *not* disappear as $\rho \rightarrow 0$. When $\rho = 0$ and $T = 10$, the asymptotic bias is exactly -0.10 in the simple model.
- **Exogenous variables make the problem worse.** Adding strictly exogenous variables does not eliminate the lagged-dependent-variable bias. In fact, the absolute bias in $\hat{\rho}$ increases because the denominator is reduced after projecting the lagged dependent variable on the exogenous regressors.
- **Bias in other coefficients.** Coefficients on exogenous variables can also become biased because their within variation may be related to the lagged dependent variable. If an exogenous regressor is positively related to the demeaned lagged dependent variable, its coefficient is pushed upward when $\hat{\rho}$ is biased downward.
- **Contribution.** The paper turns earlier Monte Carlo evidence from Nerlove and Maddala into closed-form analytical expressions, showing why their simulation results arise.
- **Empirical takeaway.** In panels with many units and short time series, standard fixed-effects OLS should not be used naively in dynamic models. The bias is structural, not a small-sample accident.

2 Conceptual setting and notation

2.1 Why panel data create a fixed-versus-random-effects choice

Panel data models often write the disturbance for individual i at time t as

$$u_{it} = f_i + \mu_t + \varepsilon_{it}, \tag{1}$$

where f_i is an individual effect, μ_t is a time effect, and ε_{it} is an idiosyncratic shock.

- If f_i is uncorrelated with regressors, random effects can exploit both between-individual and within-individual variation.
- If f_i is correlated with regressors, treating it as random and uncorrelated creates omitted-variable bias.
- The fixed-effects estimator removes f_i by using deviations from individual-specific time means.

Interpretation: The paper starts from the idea that in many economic applications f_i is not an innocuous random draw. It may represent ability, preferences, management quality, technology, or other persistent attributes. If those attributes are correlated with regressors, fixed effects are the natural approach.

2.2 Why dynamic fixed effects are different from static fixed effects

In a static fixed-effects model with strictly exogenous regressors,

$$y_{it} = x'_{it}\beta + f_i + \varepsilon_{it},$$

within demeaning gives

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i),$$

and strict exogeneity gives consistency as $N \rightarrow \infty$ for fixed T .

In a dynamic model,

$$y_{it} = \rho y_{i,t-1} + x'_{it}\beta + f_i + \varepsilon_{it},$$

within demeaning gives

$$y_{it} - \bar{y}_i = \rho(y_{i,t-1} - \bar{y}_{i,-1}) + (x_{it} - \bar{x}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

The problem is that $y_{i,t-1}$ contains past shocks, and $\bar{\varepsilon}_i$ contains those same shocks. Thus

$$\mathbb{E}[(y_{i,t-1} - \bar{y}_{i,-1})(\varepsilon_{it} - \bar{\varepsilon}_i)] \neq 0$$

for fixed T .

Interpretation: The fixed effect is removed, but dynamic endogeneity remains. This is not caused by a bad instrument, missing controls, or finite- N sampling noise. It is a fixed- T asymptotic problem.

2.3 What asymptotic sequence the paper studies

The paper studies the panel-data asymptotic sequence

$$N \rightarrow \infty, \quad T \text{ fixed.}$$

This is the relevant approximation for many micro panels: thousands of individuals, firms, or households, but only a few annual or quarterly observations.

Interpretation: The bias disappears as $T \rightarrow \infty$, but that is not helpful when T is small. The paper therefore computes the limit of the fixed-effects estimator as $N \rightarrow \infty$ while T remains fixed.

3 Models in the paper

3.1 Lagged dependent variable model

The first model is

$$y_{it} = \beta + \rho y_{i,t-1} + \sum_j \beta_j x_{ij,t} + f_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (2)$$

- ρ is the autoregressive parameter of interest.
- $x_{ij,t}$ are exogenous regressors.
- f_i is an individual fixed effect.
- ε_{it} is the idiosyncratic disturbance.

Interpretation: This model is the direct dynamic panel model. The regressor $y_{i,t-1}$ makes the fixed-effects estimator problematic because it is a function of past disturbances.

3.2 Autocorrelated residual model

The second model is

$$y_{it} = \beta + \sum_j \beta_j x_{ijt} + f_i + u_{it}, \quad (3)$$

$$u_{it} = \rho u_{i,t-1} + \varepsilon_{it}. \quad (4)$$

Rearranging the residual-autocorrelation model produces a lagged-dependent-variable representation:

$$y_{it} = \beta(1 - \rho) + \rho y_{i,t-1} + \sum_j \beta_j (x_{ijt} - \rho x_{ij,t-1}) + f_i(1 - \rho) + \varepsilon_{it}. \quad (5)$$

Interpretation: The bias results apply both when the model explicitly contains $y_{i,t-1}$ and when residual autocorrelation can be rewritten as a lagged-dependent-variable model. This is why the paper begins with both formulations.

3.3 Assumptions used in the derivation

The paper assumes:

- f_i are fixed parameters;
- $|\rho| < 1$, so the AR(1) process is stationary and ergodic;
- ε_{it} is independently distributed with mean zero and variance σ_ε^2 ;
- $\mathbb{E}_i \varepsilon_{it} = 0$ and $\mathbb{E}_i \varepsilon_{it} f_i = 0$, where $\mathbb{E}_i$ denotes an expectation across individuals at a fixed time period.

Interpretation: The result does not depend on normality in any deep way. The key ingredients are stationarity, fixed T , and the mechanical correlation created by demeaning.

4 Within transformation and source of bias

4.1 Within transformation

For any variable z_{it} , define

$$\bar{z}_i = \frac{1}{T} \sum_{t=1}^T z_{it}.$$

For the lagged dependent variable, define

$$\bar{y}_{i,-1} = \frac{1}{T} \sum_{t=1}^T y_{i,t-1}.$$

After removing f_i , the lagged-dependent-variable model becomes

$$y_{it} - \bar{y}_i = \rho(y_{i,t-1} - \bar{y}_{i,-1}) + \sum_j \beta_j (x_{ijt} - \bar{x}_{ij}) + (\varepsilon_{it} - \bar{\varepsilon}_i). \quad (6)$$

Interpretation: The fixed effect disappears, but the lagged dependent variable is no longer orthogonal to the transformed error. This is the whole source of the Nickell bias.

4.2 The simple AR(1) fixed-effects model

To isolate the bias, remove the exogenous variables. Then

$$y_{it} - \bar{y}_i = \rho(y_{i,t-1} - \bar{y}_{i,-1}) + (\varepsilon_{it} - \bar{\varepsilon}_i). \quad (7)$$

The fixed-effects estimator based on one cross-section t is

$$\hat{\rho}_t = \frac{\sum_{i=1}^N (y_{i,t-1} - \bar{y}_{i,-1})(y_{it} - \bar{y}_i)}{\sum_{i=1}^N (y_{i,t-1} - \bar{y}_{i,-1})^2}. \quad (8)$$

Using the transformed model,

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho) = \frac{A_t}{B_t}, \quad (9)$$

where

$$A_t = \mathbb{E}_i[(y_{i,t-1} - \bar{y}_{i,-1})(\varepsilon_{it} - \bar{\varepsilon}_i)], \quad (10)$$

$$B_t = \mathbb{E}_i[(y_{i,t-1} - \bar{y}_{i,-1})^2]. \quad (11)$$

Interpretation: The numerator A_t is not zero because $y_{i,t-1}$ is correlated with the average error $\bar{\varepsilon}_i$. A lagged dependent variable contains earlier shocks, and the average error contains those shocks too.

4.3 Stationary representation

Ignoring exogenous variables and time effects, stationarity gives

$$y_{it} = \frac{\beta + f_i}{1 - \rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{i,t-j}. \quad (12)$$

This representation makes the bias transparent:

- $y_{i,t-1}$ contains $\varepsilon_{i,t-1}, \varepsilon_{i,t-2}, \dots$;
- $\bar{\varepsilon}_i$ contains $\varepsilon_{i,1}, \dots, \varepsilon_{i,T}$;
- therefore $y_{i,t-1}$ and $\bar{\varepsilon}_i$ overlap in shocks whenever $t > 1$;
- demeaning subtracts $\bar{\varepsilon}_i$ from ε_{it} , so the transformed error contains a component negatively correlated with the transformed lag.

Interpretation: The sign is negative for positive ρ . Intuitively, high past y usually comes from high past shocks; those same shocks raise $\bar{\varepsilon}_i$; subtracting $\bar{\varepsilon}_i$ makes the transformed current error lower, creating a negative correlation.

5 Main bias formulas

5.1 Bias of the one-period estimator

The paper derives explicit formulas for A_t and B_t , then obtains

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho) = \frac{A_t}{B_t}. \quad (13)$$

- The bias varies with t because the overlap between $y_{i,t-1}$ and $\bar{\varepsilon}_i$ is different near the beginning, middle, and end of the sample.
- The bias is smaller at the ends of the sample period.
- It is largest around the middle of the panel.

Interpretation: A single cross-section after demeaning is already biased. Pooling over all periods averages these time-specific biases.

5.2 Bias of the pooled within estimator

The practically relevant estimator pools all time periods:

$$\hat{\rho} = \frac{\sum_{t=1}^T \sum_{i=1}^N (y_{i,t-1} - \bar{y}_{i,-1})(y_{it} - \bar{y}_i)}{\sum_{t=1}^T \sum_{i=1}^N (y_{i,t-1} - \bar{y}_{i,-1})^2}. \quad (14)$$

A convenient equivalent expression for the fixed- T asymptotic bias in the simple AR(1) model is

$$\text{plim}_{N \rightarrow \infty}(\hat{\rho} - \rho) = - \frac{\frac{1 - \rho^2}{T} \sum_{m=0}^{T-2} (T - 1 - m) \rho^m}{(T - 1) - \frac{2}{T} \sum_{m=1}^{T-1} (T - m) \rho^m}. \quad (15)$$

For reasonably large T , this simplifies to

$$\text{plim}_{N \rightarrow \infty}(\hat{\rho} - \rho) \simeq - \frac{1 + \rho}{T - 1}. \quad (16)$$

Interpretation: The leading-order approximation is the result most often remembered. It says that the bias is roughly proportional to persistence and inversely proportional to the number of time periods.

5.3 Numerical implications

Using the exact expression above:

- If $T = 10$ and $\rho = 0$, the bias is -0.1000 .
- If $T = 10$ and $\rho = 0.1$, the estimate converges to about -0.0111 , so the bias is about -0.1111 .
- If $T = 10$ and $\rho = 0.5$, the estimate converges to about 0.3378 , so the bias is about -0.1622 .
- If $T = 10$ and $\rho = 0.9$, the estimate converges to about 0.6568 , so the bias is about -0.2432 .

Interpretation: These numbers are striking because they arise even when N is arbitrarily large. More cross-sectional observations do not fix the problem when T is held fixed.

5.4 Why the bias does not disappear at $\rho = 0$

When $\rho = 0$, the dependent variable is not dynamically persistent in the population. But the within estimator still uses $y_{i,t-1} - \bar{y}_{i,-1}$ as a regressor. Since $\bar{y}_{i,-1}$ and $\bar{\varepsilon}_i$ share shocks, the transformed lag and transformed error remain correlated.

Interpretation: The dynamic fixed-effects bias is not only a high-persistence problem. Persistence makes it worse, but the bias is already present in a short panel because of the within transformation.

5.5 Connection to the Hurwicz bias

The paper compares the dynamic panel bias with the classic Hurwicz bias in a single time-series AR(1) with an estimated intercept. The Hurwicz bias is larger to order $1/T$:

$$\text{Hurwicz bias} \approx - \frac{1 + 3\rho}{T - 1}. \quad (17)$$

The difference is that the dynamic panel limit studies

$$\frac{\mathbb{E}(A)}{\mathbb{E}(B)},$$

whereas the single-time-series bias involves

$$\mathbb{E}\left(\frac{A}{B}\right).$$

Interpretation: Single time-series bias has additional terms from the covariance of numerator and denominator and the variance of the denominator. The panel bias is usually smaller than the Hurwicz bias, but it matters more in many applications because micro panels often have very short T .

6 Including exogenous variables

6.1 Stacked within model

With exogenous variables, write the transformed model for each t as

$$y_t = \rho y_{t-1} + X_t b + \varepsilon_t, \quad (18)$$

where all variables are understood to be demeaned within individual. Stacking over $t = 1, \dots, T$ gives

$$y = \rho y_{-1} + Xb + \varepsilon. \quad (19)$$

Let

$$M = I - X(X'X)^{-1}X' \quad (20)$$

be the residual-maker matrix that projects out the exogenous variables. Then the within estimator for ρ is the coefficient on y_{-1} after partialling out X .

Interpretation: Adding exogenous regressors means that the estimator uses only the part of the demeaned lagged dependent variable not explained by the demeaned exogenous regressors.

6.2 Bias in *hatrho* with exogenous variables

The paper shows that

$$\text{plim}_{N \rightarrow \infty}(\hat{\rho} - \rho) = \frac{\text{plim}_{N \rightarrow \infty} \frac{1}{NT} y'_{-1} M \varepsilon}{\text{plim}_{N \rightarrow \infty} \frac{1}{NT} y'_{-1} M y_{-1}}. \quad (21)$$

Since X is exogenous, projecting out X does not change the numerator. But it reduces the denominator because

$$y'_{-1} M y_{-1} \leq y'_{-1} y_{-1}.$$

Therefore, for $\rho > 0$, the bias in $\hat{\rho}$ remains negative and becomes larger in absolute value.

Interpretation: Exogenous controls do not solve Nickell bias. They can make the dynamic coefficient more biased because they remove variation from the lagged dependent variable without removing the correlation between the transformed lag and the transformed error.

6.3 Bias in coefficients on exogenous variables

Let $\hat{b}$ be the OLS estimator for the exogenous-variable coefficients after within transformation. The paper gives the relationship

$$\text{plim}_{N \rightarrow \infty}(\hat{b} - b) = -\text{plim}_{N \rightarrow \infty} [(X'X)^{-1} X' y_{-1}] \text{plim}_{N \rightarrow \infty}(\hat{\rho} - \rho). \quad (22)$$

- If an exogenous regressor is positively related to the demeaned lagged dependent variable, then $X'y_{-1} > 0$.
- If $\hat{\rho} - \rho < 0$, the product above implies an upward bias in the coefficient on that regressor.
- If the regressor is negatively related to the demeaned lagged dependent variable, the sign reverses.

Interpretation: The bias is not isolated to the lagged dependent variable. It spills over into other coefficients through the correlation structure among regressors.

6.4 Why this matters for empirical work

Suppose an applied researcher estimates

$$y_{it} = \rho y_{i,t-1} + \beta x_{it} + f_i + \varepsilon_{it}$$

with firm or worker fixed effects. If x_{it} is correlated with past y , then a downward-biased estimate of ρ forces the regression to attribute too much or too little of the remaining variation to x_{it} .

Interpretation: In dynamic panel regressions, bias in the persistence parameter can contaminate substantive coefficients. The problem is especially acute when regressors are themselves persistent.

7 Monte Carlo comparisons and validation

7.1 Comparison with Nerlove

Nerlove's Monte Carlo experiments used panels with $T = 10$ and $N = 25$ for several values of ρ . Nickell reports that the analytical formula closely matches the Monte Carlo estimates:

True ρ	Nerlove average estimate	Nickell formula estimate
0.0	-0.10083	-0.10000
0.1	-0.01115	-0.01108
0.5	0.33354	0.33779
0.9	approximately 0.657 under lower fixed-effect variance	0.65677

Interpretation: The formula is asymptotic in N , yet it works well in simulations with only $N = 25$. This confirms that the bias is primarily a fixed- T phenomenon rather than a finite- N artifact.

7.2 Comparison with Maddala

Maddala considers a case with $\rho = 0.7$ and a nominal $T = 10$ experiment that corresponds to $T = 9$ in Nickell's notation. Without an exogenous variable, Maddala obtains an estimate around 0.475, while the analytical formula gives 0.4805. With an exogenous variable whose true coefficient is 0.5, Maddala obtains a much lower estimate of ρ around 0.3178 and an upward-biased estimate of the exogenous coefficient.

Interpretation: This comparison illustrates the paper's exogenous-variable result: adding a correlated exogenous regressor can worsen the downward bias in $\hat{\rho}$ and distort the coefficient on the regressor.

8 Visual notes: original equation excerpts

This paper has no empirical tables or plotted figures. The visuals below are compact groupings of the key original equation and result excerpts, directly cropped from the paper.

8.1 Paper header, abstract, and core dynamic models

BIASES IN DYNAMIC MODELS WITH FIXED EFFECTS

BY STEPHEN NICKELL¹

It is well known from the Monte-Carlo work of Nerlove that using the standard within-group estimator for dynamic models with fixed individual effects generates estimates which are inconsistent as the number of “individuals” tends to infinity if the number of time periods is kept fixed. In this paper we present analytical expressions for these inconsistencies for the first order autoregressive case.

(a) Paper identity and abstract.

Read:

- The abstract states the paper’s contribution: analytical expressions for inconsistencies in first-order autoregressive dynamic fixed-effects models.
- The model block shows both the lagged dependent variable model and the autocorrelated residual model.
- The residual-autocorrelation specification can be rewritten as a lagged-dependent-variable model, so both lead to the same fixed-effects bias issue.

Economic message: The paper is short but foundational because it identifies a precise econometric failure mode in a workhorse panel estimator.

8.2 Within transformation and the first bias setup

Rearranging (3) and (4) gives

$$(5) \quad y_{it} = \beta(1 - \rho) + \rho y_{it-1} + \sum_j \beta_j (x_{ijt} - \rho x_{ijt-1}) + f_i(1 - \rho) + \epsilon_{it}.$$

Concentrating our exposition on the lagged dependent variable model (2), the standard estimation procedure is to start by eliminating the fixed effects f_i . This may be done in any number of ways but the standard technique is to subtract the time mean of (2) from (2) itself to yield

$$(6) \quad y_{it} - y_{i\cdot} = \rho(y_{it-1} - y_{i\cdot-1}) + \sum_j \beta_j (x_{ijt} - x_{ij\cdot}) + (\epsilon_{it} - \epsilon_{i\cdot})$$

where for any variable z_{it} , $z_{i\cdot} = (1/T) \sum_{t=1}^T z_{it}$ and $z_{i\cdot-1} = (1/T) \sum_{t=0}^{T-1} z_{it}$. It is clear that OLS estimates based on (6) will be biased *even* if N , the number of individuals, goes to infinity. This arises because in these circumstances the correlation between y_{it-1} and ϵ_{it} , for example, does not go to zero. The remainder of the paper is devoted to an analysis of these biases and is set out as follows. In the next section we shall compute the bias as $N \rightarrow \infty$ in the model with no exogenous variables and we shall then look at the effect of including exogenous variables on these results. In subsequent sections we compare our analytical computations with some of the extensive Monte-Carlo results presented in Nerlove [12, 13] and Maddala [9].

1. BIASES IN AUTOREGRESSIVE MODEL WITH FIXED EFFECTS

If we remove the exogenous variables from (6), we have

$$(7) \quad y_{it} - y_{i\cdot} = \rho(y_{it-1} - y_{i\cdot-1}) + (\epsilon_{it} - \epsilon_{i\cdot}) \quad (i = 1, \dots, n; t = 1 \dots T).$$

Note that this equation follows from both the lagged dependent variable and the residual autoregression model. In order to estimate ρ we have a number of options. The standard method is to use OLS on (7) pooling all the cross-sections. It is perfectly legitimate, however, to use directly only one cross-section in estimating (7) although the data on all the others have, of course, already been used to compute the time means. If we use the t th cross-section we may define an OLS estimate

$$(8) \quad \hat{\rho}_t = \frac{\sum_{i=1}^N (y_{it-1} - y_{i\cdot-1})(y_{it} - y_{i\cdot})}{\sum_{i=1}^N (y_{it-1} - y_{i\cdot-1})^2}.$$

(a) Equations (5)–(8): transformation and estimator.

for the first-order autoregressive case. Two models will be considered. These are, omitting the time effects for simplicity of exposition,

$$(2) \quad y_{it} = \beta + \rho y_{it-1} + \sum_j \beta_j x_{ijt} + f_i + \epsilon_{it} \quad (i = 1 \dots N; t = 1 \dots T),$$

and

$$(3) \quad y_{it} = \beta + \sum_j \beta_j x_{ijt} + f_i + u_{it} \quad (i = 1 \dots N; t = 1 \dots T),$$

$$(4) \quad u_{it} = \rho u_{it-1} + \epsilon_{it}.$$

(b) Equations (2)–(4): two dynamic models.

limits as $N \rightarrow \infty$. Thus we have, using (7),

$$(9) \quad \text{plim}_{N \rightarrow \infty} \hat{\rho}_t = \rho + \frac{\text{plim}_{N \rightarrow \infty} 1/N \sum_{i=1}^N (y_{it-1} - y_{i\cdot-1})(\epsilon_{it} - \epsilon_{i\cdot})}{\text{plim}_{N \rightarrow \infty} 1/N \sum_{i=1}^N (y_{it-1} - y_{i\cdot-1})^2}$$

or

$$(10) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho) = A_t / B_t, \quad \text{say.}$$

To avoid continual use of more complicated terminology we shall, in future generally refer to anything of the form $\text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho)$ as a bias. Since ϵ_{it} are random drawings from a normal distribution when t is fixed we can replace plims as $N \rightarrow \infty$ by expectations across i , E_i , and thus obtain

$$A_t = E_i(y_{it-1} - y_{i\cdot-1})(\epsilon_{it} - \epsilon_{i\cdot})$$

or

$$(11) \quad A_t = -E_i y_{it-1} \epsilon_{it} - E_i y_{i\cdot-1} \epsilon_{it} + E_i y_{i\cdot-1} \epsilon_{i\cdot},$$

noting that $E_i y_{it-1} \epsilon_{it} = 0$. Before proceeding it is worth pointing out that stationarity implies the following result. Removing the exogenous variables and time effects, (2) implies

$$(12) \quad y_{it} = (\beta + f_i)/(1 - \rho) + \sum_{j=0}^{\infty} \rho^j \epsilon_{it-j}.$$

Then (11) and (12) imply

$$\begin{aligned} A_t = & -E_i \left\{ \left(\sum_{j=0}^{\infty} \epsilon_{it-j-1} \rho^j \right) \left(\frac{1}{T} \sum_{s=1}^T \epsilon_{is} \right) \right\} - E_i \left\{ \frac{\epsilon_{it}}{T} \sum_{s=1}^T \sum_{j=0}^{\infty} \epsilon_{is-j-1} \rho^j \right\} \\ & + E_i \left\{ \left(\frac{1}{T} \sum_{s=1}^T \sum_{j=0}^{\infty} \epsilon_{is-j-1} \rho^j \right) \left(\frac{1}{T} \sum_{s=1}^T \epsilon_{is} \right) \right\} \end{aligned}$$

where we have used the fact that $E_i \epsilon_{it} = E_i \epsilon_{it} f_i = 0$. Noting that $E_i \epsilon_{it}^2 = \sigma^2$ independent of t , we have after some manipulation

$$A_t = -\frac{\sigma^2}{T} \frac{(1 - \rho^{t-1})}{1 - \rho} - \frac{\sigma^2}{T} \frac{(1 - \rho^{T-t})}{1 - \rho} + \frac{\sigma^2}{T} \left[\frac{1}{1 - \rho} - \frac{1}{T} \frac{(1 - \rho^T)}{(1 - \rho)^2} \right]$$

where the three terms correspond to those in the previous section. Collecting terms then yields

$$(13) \quad A_t = -\frac{\sigma^2}{T(1 - \rho)} \left\{ 1 - \rho^{t-1} - \rho^{T-t} + \frac{1}{T} \frac{(1 - \rho^T)}{(1 - \rho)} \right\}.$$

(b) Equations (9)–(13): numerator of the bias.

Read:

- The left excerpt shows the within transformation and the one-period OLS estimator.
- The right excerpt starts from the probability limit of the estimator and derives the numerator A_t .
- Stationarity is used to express y_{it} as an infinite weighted sum of shocks.

Economic message: The key object is the covariance between the transformed lag and transformed error. Demeaning creates that covariance mechanically.

8.3 Cross-section-specific bias and pooled bias

$$(14) \quad B_t = \frac{\sigma_\epsilon^2}{1-\rho^2} \left(1 - \frac{1}{T}\right) + \frac{2\rho}{1-\rho^2} A_t.$$

The final value for the bias is thus given by

$$(15) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho) = \left\{ \frac{2\rho}{1-\rho^2} - \left[\frac{1+\rho}{T-1} \left(1 - \rho^{t-1} - \rho^{T-t} + \frac{1}{T} \frac{(1-\rho^T)}{(1-\rho)} \right) \right]^{-1} \right\}^{-1}$$

or

$$(16) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho}_t - \rho) = \frac{-(1+\rho)}{T-1} \left\{ 1 - \rho^{t-1} - \rho^{T-t} + \frac{1}{T} \frac{(1-\rho^T)}{1-\rho} \right. \\ \left. \times \left[1 - \frac{2\rho}{(T-1)(1-\rho)} \left[1 - \rho^{t-1} - \rho^{T-t} + \frac{1}{T} \frac{(1-\rho^T)}{1-\rho} \right] \right] \right\}^{-1}$$

(a) Equations (14)–(16): cross-section-specific bias.

$$\hat{\rho} = \frac{\sum_{t=1}^T \sum_{i=1}^N (y_{it} - y_{i, \cdot -1})(y_{it} - y_{i, \cdot})}{\sum_{t=1}^T \sum_{i=1}^N (y_{it-1} - y_{i, \cdot -1})^2}$$

and in our standard notation the bias is given by

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) = \sum_{t=1}^T A_t / \sum_{t=1}^T B_t$$

which yields

$$(17) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) = \left\{ \frac{2\rho}{1-\rho^2} - \left[\frac{1+\rho}{T-1} \left(1 - \frac{1}{T} \frac{(1-\rho^T)}{(1-\rho)} \right) \right]^{-1} \right\}^{-1}$$

or

$$(18) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) = \frac{-(1+\rho)}{T-1} \left\{ 1 - \frac{1}{T} \frac{(1-\rho^T)}{1-\rho} \right\} \\ \times \left\{ 1 - \frac{2\rho}{(1-\rho)(T-1)} \left[1 - \frac{1}{T} \frac{(1-\rho^T)}{1-\rho} \right] \right\}^{-1}$$

Furthermore, for reasonably large values of T we have the simple approx

$$(19) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) \simeq \frac{-(1+\rho)}{T-1}.$$

On the other hand for small values of T we have

$$\text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) = \frac{-(1+\rho)}{2} \quad \text{for } T = 2, \\ = \frac{-(2+\rho)(1+\rho)}{2} \quad \text{for } T = 3,$$

(b) Equations (17)–(19): pooled within-estimator bias.

Read:

- The cross-section-specific expression depends on t .
- The pooled estimator averages across all time periods.
- The most useful approximation is $-(1+\rho)/(T-1)$.

Economic message: The bias is not small when T is small. It is negative for positive ρ and shrinks only as the time dimension grows.

8.4 Hurwicz comparison and exogenous-variable setup

which can hardly be ignored. Furthermore, the bias does not go to zero as ρ goes to zero. However, these biases are not as severe as the standard Hurwicz biases associated with first-order autoregressive processes with a constant term. In this case, at least to order $1/T$, the bias is $-(1+3\rho)/T - 1$ which is larger than those considered above. This approximation and a large number of related results may be found in Mariott and Pope [10] and Kendall [7]. The standard Hurwicz bias differs from that given in (17) basically because if we let $A = \sum_i (y_{it-1} - y_{i,-1})(\epsilon_{it} - \epsilon_{i,-1})$ and $B = \sum_i (y_{it-1} - y_{i,-1})^2$, the Hurwicz bias in the standard regression (with one value of i) is given by $E(A/B)$. We, of course, are computing the bias as $N \rightarrow \infty$ and are thus considering $E(A)/E(B)$ where expectations are all taken across i . These expressions are related in the sense that the approximation to the standard Hurwicz bias to order T^{-1} is given by

$$E(A/B) = \frac{E(A)}{E(B)} \left\{ 1 - \frac{\text{cov}(AB)}{E(A)E(B)} + \frac{\text{var}(B)}{E^2(B)} \right\}$$

(a) Comparison with Hurwicz bias.

2. THE INCLUSION OF EXOGENOUS VARIABLES

In this section we shall concentrate on model (2) with the lagged endogenous variable. If we define the following matrices

$$\begin{aligned}\tilde{y}_t &= [y_{it} - y_{i,-}], \quad N \times 1 \text{ vector,} \\ \tilde{y}_{t-1} &= [y_{it-1} - y_{i,-1}], \quad N \times 1 \text{ vector,} \\ \tilde{X}_t &= [x_{ijt} - x_{ij,-}], \quad N \times J \text{ matrix,} \\ \tilde{\epsilon}_t &= [\epsilon_{it} - \epsilon_{i,-}], \quad N \times 1 \text{ vector,} \\ b &= [\beta_j], \quad J \times 1 \text{ vector,}\end{aligned}$$

(6) may then be written in deviation form as

$$(20) \quad \tilde{y}_t = \rho \tilde{y}_{t-1} + \tilde{X}_t b + \tilde{\epsilon}_t$$

where we have introduced J exogenous variables. To obtain the most efficient estimates we may now stack these equations over the T time periods to obtain

$$(21) \quad \begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_T \end{bmatrix} = \rho \begin{bmatrix} \tilde{y}_0 \\ \tilde{y}_1 \\ \vdots \\ \tilde{y}_{T-1} \end{bmatrix} + \begin{bmatrix} \tilde{X}_1 \\ \tilde{X}_2 \\ \vdots \\ \tilde{X}_T \end{bmatrix} b + \begin{bmatrix} \tilde{\epsilon}_1 \\ \tilde{\epsilon}_2 \\ \vdots \\ \tilde{\epsilon}_T \end{bmatrix}$$

(b) Equations (20)–(21): adding exogenous variables.

Read:

- The left excerpt compares the panel bias with the single-time-series Hurwicz bias.
- The right excerpt introduces stacked notation and the projection matrix M .
- The presence of M shows that the lagged dependent variable is residualized on exogenous regressors before estimating ρ .

Economic message: Adding controls does not remove the source of the bias. It changes the denominator and can increase the absolute bias.

8.5 Direction of exogenous-variable bias and Monte Carlo comparison

or

$$(22) \quad \tilde{y} = \rho \tilde{y}_{-1} + \tilde{X}b + \tilde{\epsilon}$$

again in obvious notation. If $\hat{\rho}$, $\hat{b}$ are the OLS estimates, using standard procedures we obtain

$$(23) \quad \hat{\rho} - \rho = (\tilde{y}_{-1}'M\tilde{y}_{-1})^{-1}\tilde{y}_{-1}'M\tilde{\epsilon},$$

$$(24) \quad \hat{b} - b = -(\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y}_{-1}(\hat{\rho} - \rho) + (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{\epsilon},$$

where $M = I - \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}'$. Taking plims as $N \rightarrow \infty$ and noting that

$$\text{plim}_{N \rightarrow \infty} \frac{1}{NT} \tilde{y}_{-1}'M\tilde{\epsilon} = \text{plim}_{N \rightarrow \infty} \frac{1}{NT} \tilde{y}_{-1}'\tilde{\epsilon}$$

since $\tilde{X}$ is exogenous, we have

$$(25) \quad \text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho) = \left(\text{plim}_{N \rightarrow \infty} \frac{1}{NT} \tilde{y}_{-1}'M\tilde{y}_{-1} \right)^{-1} \text{plim}_{N \rightarrow \infty} \frac{1}{NT} \tilde{y}_{-1}'\tilde{\epsilon}$$

and

$$(26) \quad \text{plim}_{N \rightarrow \infty} (\hat{b} - b) = - \text{plim}_{N \rightarrow \infty} [(\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y}_{-1}] \text{plim}_{N \rightarrow \infty} (\hat{\rho} - \rho).$$

Note first from our previous analysis that we have already calculated $\text{plim}_{N \rightarrow \infty} (1/NT)\tilde{y}_{-1}'\tilde{\epsilon}$ and this is given by

$$(27) \quad \text{plim}_{N \rightarrow \infty} \frac{1}{NT} \tilde{y}_{-1}'\tilde{\epsilon} = \frac{1}{T} \sum A_t = \frac{-\sigma_\epsilon^2}{T(1-\rho)} \left(1 - \frac{1}{T} \frac{(1-\rho^T)}{1-\rho} \right).$$

This result remains unaltered by the introduction of exogenous variables since their incorporation into equation (12) will have no effect given they are uncorrelated with the error term. When ρ is positive we may derive the direction of the biases. Since A_t is negative, $\text{plim}(\hat{\rho} - \rho)$ must be negative and will indeed be larger (in absolute value) than if the exogenous variables are omitted since the denominator of the expression in (25) is reduced by the inclusion of M . The bias on $\hat{b}$ depends on the relationship between the exogenous variables and $\tilde{y}_{-1}$. If an exogenous variable is positively related (in the regression sense) to $\tilde{y}_{-1}$, then (26) indicates that its coefficient will be upward biased and vice-versa.

(a) Direction of bias with exogenous variables.

different values of the variance of f_i relative to that of ϵ_{it} , a number which does not influence the asymptotic bias and indeed, as he himself points out, does not affect his $\hat{\rho}$ estimates except when $\rho = 0.9$. For $\rho = 0.0, 0.1, 0.5$ the average Nerlove Monte Carlo estimates of $\hat{\rho}$ reported in Table C1 are $-0.10083, -0.01115$, and 0.33354 respectively.⁸ The corresponding $\hat{\rho}$'s given by equation (17) are $-0.10000, -0.01108$, and 0.33779 . They are thus more or less exact even though they are only asymptotic in N and Nerlove has $N = 25$. The results for $\rho = 0.9$ are not quite so clear cut. Equation (17) gives $\hat{\rho} = 0.65677$ whereas the Monte Carlo results only yield this value if the variance of f_i is not large relative to the total error variance. Thus the average Monte Carlo estimate of $\hat{\rho}$ is 0.657 if we only consider those experiments where the variance of f_i is less than one third of the total—otherwise the Monte Carlo estimates become considerably higher and our asymptotic result is no longer accurate.

Turning to the introduction of exogenous variables we may note an experiment by Maddala for $\rho = 0.7$ where he again has $N = 25$ and although he claims to have $T = 10$ it appears that he only generates ten values of the dependent variable for each i , which would imply $T = 9$ in our notation given the lagged value on the right-hand side. When there is no exogenous variable he generates $\hat{\rho} = 0.475$ (equation (17) yields $\rho = 0.4805$) whereas the introduction of an exogenous variable with a true coefficient equal to 0.5 reduces $\hat{\rho}$ to 0.3178 and generates an estimate of β which is strongly upward biased as we might expect from (26).

SUMMARY

We have presented analytical expressions for the asymptotic biases in first-order autoregressive models estimated by OLS using panel data and including individual fixed effects. These asymptotic biases are shown to be both large and to coincide almost exactly with the estimates provided by the Monte Carlo studies of Nerlove (1967) and Maddala (1971).

(b) Monte Carlo comparison and summary.

Read:

- The left excerpt explains why the bias in $\hat{\rho}$ becomes more negative after including exogenous variables and how this affects $\hat{b}$.
- The right excerpt compares the analytical result with Nerlove and Maddala simulations.
- The simulations closely match the formulas.

Economic message: The analytical bias formula is not merely theoretical. It explains simulation evidence almost exactly.

9 How to use the paper in empirical work

9.1 What not to do

Do not interpret fixed-effects OLS estimates of

$$y_{it} = \rho y_{i,t-1} + x'_{it}\beta + f_i + \varepsilon_{it}$$

as consistent when T is short and N is large. In this setting, increasing the cross-sectional sample size only makes the estimator converge more precisely to a biased limit.

Interpretation: Large samples do not solve all problems. The relevant asymptotic dimension is T , not N .

9.2 When the bias is likely severe

The bias is likely severe when:

- T is small;
- ρ is positive and large;
- regressors are persistent and correlated with the lagged dependent variable;
- the empirical question depends on accurately estimating persistence;
- the model includes fixed effects and lagged outcomes simultaneously.

Interpretation: In many labor, industrial organization, macro-finance, household, and firm panels, T is exactly the dimension that is small. This is why the result became central to dynamic panel econometrics.

9.3 Conceptual implication for later estimators

Although this paper does not develop later GMM estimators, its result motivates methods that avoid the within-estimator bias, such as instrumental-variable approaches using deeper lags or transformations that exploit moment conditions.

Interpretation: The paper provides the diagnostic problem. Later dynamic panel estimators are responses to this problem.

10 Short summary

- Nickell studies dynamic panel models with individual fixed effects.
- Fixed effects are important because unobserved individual heterogeneity is often correlated with regressors.
- In static panels with strictly exogenous regressors, within transformation can produce consistent estimates for fixed T and large N .
- In dynamic panels, within transformation creates correlation between the demeaned lagged dependent variable and the demeaned error term.
- This correlation does not vanish as $N \rightarrow \infty$ when T is fixed.
- The asymptotic bias in the AR(1) coefficient is negative for positive ρ .
- The leading approximation is $-(1 + \rho)/(T - 1)$.
- The bias can be large even when $T = 10$.
- The bias remains even when $\rho = 0$.
- Adding exogenous variables does not solve the problem and can make the bias in $\hat{\rho}$ larger in absolute value.
- Coefficients on exogenous variables can be biased through their relationship with the lagged dependent variable.
- The analytical formula closely matches Nerlove's and Maddala's Monte Carlo results.
- The paper is the canonical reference for why dynamic fixed-effects OLS is inconsistent in short panels.

11 Conclusion

11.1 Core conclusion

The standard within estimator in a dynamic fixed-effects panel is inconsistent when $N \rightarrow \infty$ and T is fixed. The problem arises because demeaning the data introduces a correlation between the lagged dependent variable and the transformed disturbance. This correlation remains no matter how many individuals are added to the sample.

11.2 Economic and econometric intuition

The lagged dependent variable contains past shocks. The demeaned error subtracts the individual's average shock. Because the average shock includes past shocks, the transformed lagged dependent variable and transformed error are mechanically related. With positive persistence, this relationship pushes the estimated autoregressive coefficient downward.

11.3 Takeaway

The paper's lesson is simple and powerful: in dynamic panels, fixed effects remove unobserved heterogeneity but create a new endogeneity problem. For short panels, researchers need estimators designed for dynamic panel bias rather than ordinary within-group OLS.